

Ariel Lubonja

PhD Candidate, ML Systems Performance Engineering. Graduating 05/28

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EDUCATION

Johns Hopkins University

PhD, Computer Science — GPA: 3.92/4.0 — Advisors: R. Burns, J. Vogelstein, A. Yuille

Aug 2022 – May 2028

MS, Data Science — GPA: 3.8/4.0 — Dean's List

Jan 2021 – May 2022

RESEARCH EXPERIENCE

Google YDF — Open-Source Research Collaboration

May 2025 – Present

Joint work with the Google YDF team (Zurich)

- Cut Google YDF's oblique training time 6x (**12 h to 2 h**) by **optimizing compute kernels** in 7 PRs.
- Designed two novel kernel optimizations (1.7–2.5x): AVX-512 histogram binning and cost-model split selection.
- Optimizing **memory-bound GPU kernel** with a novel 1-pass design (25x over CPU) w/ junior PhD student
- Sped up gradient-boosted trees on high-dimensional data **50x by perf profiling** and fixing a mutex bottleneck.
- Designing a **distributed training** GBT method that minimizes communication.

NVIDIA PhysicsNeMo Contributor

Nov 2025 – Feb 2026

2 merged pull requests

- Implemented and tested a new API in NVIDIA PhysicsNeMo (PyTorch); merged upstream in 2 pull requests.
- Improved runtime by 2.5x on low-resolution inputs by eliminating a redundant API call.
- Funded by and contributing to the Johns Hopkins Turbulence Database, the largest public turbulence archive.

LLM Inference Efficiency & Fine-Tuning

Mar 2024 – Jul 2024

Class project

- Early-exit GPT-2 decoding from intermediate layers; measured per-layer output agreement and GPU latency.
- Cross-layer attention redundancy in GPT-2: per-head Pearson correlation of attention rows between blocks.
- LoRA fine-tuning of BLOOM 560M–3B (0.28% trainable params, fp16, 8-bit inference) on T4/A100 GPUs.

Scalable Node Embedding for Graph ML

Dec 2023 – May 2024

Open-source package and IPDPS-W 2024 (GrAPL) paper

- Cut node embedding of the **billion-edge Friendster graph** from 56 min to 6.4 s on 24 cores (500x vs. Python original, 17x vs. a Numba JIT baseline) via a lock-free, edge-parallel, memory-bandwidth-bound Ligra rewrite.

GPU-Accelerated Orthogonal Matching Pursuit

May 2022 – Jan 2023

PyPI package *batched-omp*, arXiv preprint and open-source code

- Built a 310x faster drop-in GPU replacement for scikit-learn's Orthogonal Matching Pursuit (OMP).
- Used **batched cuBLAS** calls (in **PyTorch**), Gramian precomputation + Cholesky factorization.

WORK EXPERIENCE

Software Developer

May 2017 – Jan 2021

jBoxers

Sofia, Bulgaria

- NLP chatbot for USC (Mycroft, Dialogflow); Library of Congress Virtual Card Catalog (Angular, Django, Java EE).

TECHNICAL SKILLS

Languages & Tools: C/C++, Python, CUDA, Java, Bash, Linux, Git, Bazel

Performance Engineering: profiling & benchmarking (Intel VTune, Advisor, perf, NVIDIA Nsight Systems),

GPU/CPU kernel optimization, memory-hierarchy analysis, SIMD (AVX-512, Highway), OpenMP, Slurm, cuBLAS, Zarr

Machine Learning: PyTorch, JAX, Transformers, Tree Ensembles, Graph Learning, Sparse ML, LLMs

TEACHING, HONORS & LEADERSHIP

Teaching Assistant: Big Data Systems (EN.601.612), Fall 2026 — Spark, Kafka, Hadoop, NoSQL

Parallel Programming & Performance Engineering (EN.601.620), 2022–2024 — SIMD, multicore, memory, CUDA

Fellowship to Johns Hopkins University: Fulbright-style fully funded Albanian fellowship. 1 of 14 nationwide

Vice President: Upsilon Pi Epsilon (international CS honor society), Johns Hopkins chapter